AN ETALE MORPHISM OF ALGEBRAIC SPACES IS LOCALLY OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence; the proof is a two-step chain of same-chapter lemmas and is complete.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- An etale morphism of algebraic spaces is locally of finite presentation. Lemma lemma-etale-locally-finite-presentation, used at proof lines 8039-8043. Is the first step of the chain.
- A morphism locally of finite presentation is locally of finite type. Lemma lemma-finite-presentation-finite-type (5744-5748), used at proof lines 8041-8044. Is the second step of the chain.
- The definition of etaleness for morphisms of algebraic spaces. used at proof lines 8039-8040. Identifies the starting point of the chain.

Source reference. Original lemma 0468: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5463–5463. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0464: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5459–5459. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *An étale morphism of algebraic spaces is locally of finite type.*

Proof. An étale morphism is locally of finite presentation and a morphism locally of finite presentation is locally of finite type, see Lemmas 0468 and 0464. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.